

MFE 5231 (2nd Term, 2025-2026)

Case Questions

Joseph Cheng

1) The Boeing 7E7

- a) What is your estimate of the cost of capital for the 7E7 (787) project? ie. estimate the WACC for the commercial segment of Boeing. Which beta should you use (you need to use comparable firm information in Exh. 10 to estimate a levered beta for the commercial segment of Boeing)? Which risk-free rate and market risk premium should you use when using the CAPM to estimate the cost of equity for the commercial segment? What capital structure weights should you use to derive the WACC for the commercial segment (revenue or market value weights etc.)?
- b) Judge against the WACC computed in (a), will the 7E7 (787) project create value for Boeing's investors? What should Michael Bair / the board do?

Hint: $\beta_{\text{firm}} = W_{\text{com}} \times \beta_{\text{com}} + W_{\text{def}} \times \beta_{\text{def}}$

The beta that we observed is levered beta. The relationship between levered and unlevered beta is :

$$\beta^L = \beta^U \times [1 + (1-t) D/E]$$

Note that the average equity market risk premium for the past 74 years (as at Dec 31, 2002) are as follows:

	<u>Over T-Bill</u>	<u>Over T-Bond</u>
Geometric mean	6.40%	4.70%
Arithmetic mean	8.40%	6.40%

2) Star River Electronics LTD.

- a) Can the firm repay its loan within a reasonable period? Hint: computes the sustainable rate of growth and compares it with the growth rate that the firm is experiencing to determine if the firm is able to self-sustain its growth without additional borrowing.
- b) Estimate the WACC for Star River. Use comparable firm information together with unlever and re-lever of beta to obtain the beta estimate to be used in conjunction with the CAPM to compute the cost of equity. Weight the cost of equity so obtained with the cost of debt to arrive at the WACC.

- c) How should Star River deal with the packaging-machine investment? Note that the ‘invest now’ decision and the ‘wait’ alternative results in identical cash flows after 13 years.

3) Warren E. Buffett, 2005

- a) What does the stock market seem to be saying about the acquisition of PacifiCorp by Berkshire Hathaway? Specifically, what does the \$2.55 billion gain in Berkshire’s market value of equity imply about the intrinsic value of PacifiCorp?
- b) Based on the multiples for comparable regulated utilities, what is the range of possible values for PacifiCorp equity? Is Buffett paying too much to acquire PacifiCorp?
- c) What is your assessment of Berkshire’s investments in Buffett’s “Big Four” : American Express, Coca Cola, Gillette and Wells Fargo?
- d) Please critically assess Buffett’s investment philosophy and identify points where you agree and disagree with him.

4) TeleTech Corporation, 2005

- a) How does the choice of constant vs. risk-adjusted hurdle rates affect the evaluation of Tele-Tech’s two segments? What are the implications for Tele-Tech’s resource allocation strategy?
- b) Please estimate segment WACC for Tele-Tech. Note that the capital structure weights of each segments is approximately equal to the average of comparable firms in Exh. 3.
- c) If Tele-Tech began allocating capital on the basis of risk-adjusted hurdle rate, would it destroy value? Has Products and Systems destroyed value? Should Tele-Tech implement the risk-adjusted hurdle rate system?
- d) Briefly conjecture on the likely barriers for the implementation of the risk-adjusted hurdle rate system.
- e)

5) Nike, Inc. : Cost of Capital

- a) Do you agree with Joanna Cohen’s WACC calculation? Why or why not?
- b) Using the CAPM to estimate the *cost of equity*, what is your estimate of Nike’s WACC? What assumption did you make when selecting the risk-free rate from Exhibit 4?
- c) Estimate the cost of equity using the Dividend Growth Model and subsequently compute the WACC.

- d) Use the Earning Capitalization Model (ie, the reciprocal of the PE ratio) to estimate the cost of equity for Nike. Is this a reasonable estimate of the cost of equity?
- e) What is the likely range for the WACC of Nike based upon your results from (b) and (c)?

6) H. J. Heinz: Estimating the Cost of Capital in Uncertain Times

- a) What was the yield on the 2032 outstanding Heinz-debt issue as of the end of April 2010? What was the yield one year earlier?
- b) What was the WACC for Heinz at the start of fiscal year 2010? What was the WACC one year earlier? Hint: Use a 5.5% MRP and marginal tax rate of 30%.
- c) What is your best estimate of the WACC for Kraft Foods, Campbell Soup Company and Del Monte Foods? How do these Waccs influence your thought about the WACC for Heinz?